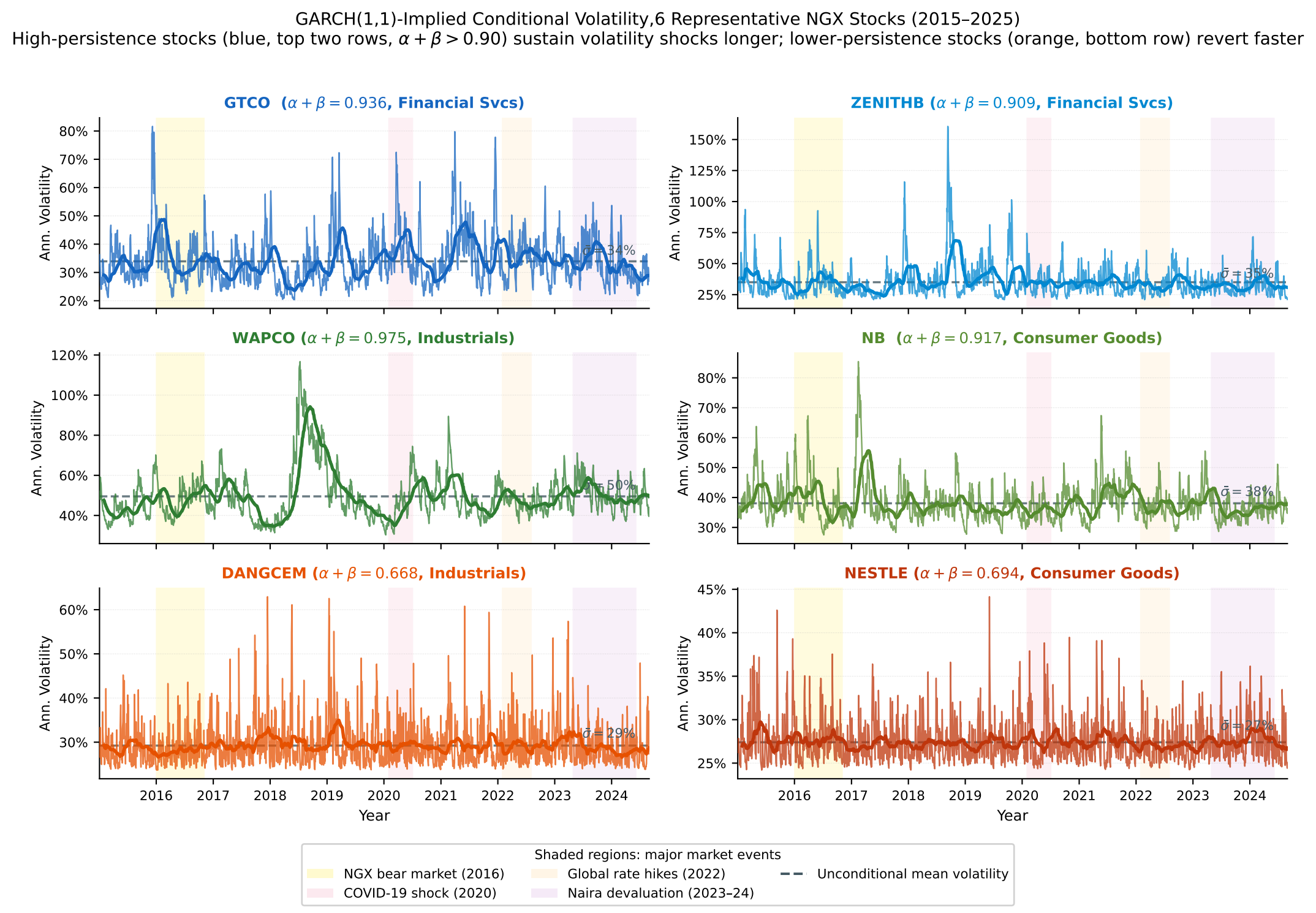

GARCH(1,1)-Implied Conditional Volatility,6 Representative NGX Stocks (2015–2025)
High-persistence stocks (blue, top two rows, $\alpha+\beta>0.90$) sustain volatility shocks longer; lower-persistence stocks (orange, bottom row) revert faster
GTCO ($\alpha+\beta=0.936$, Financial Svcs)
ZENITHB ($\alpha+\beta=0.909$, Financial Svcs)
WAPCO ($\alpha+\beta=0.975$, Industrials)
NB ($\alpha+\beta=0.917$, Consumer Goods)
DANGCEM ($\alpha+\beta=0.668$, Industrials)
NESTLE ($\alpha+\beta=0.694$, Consumer Goods)
Ann. Volatility
$\bar{\sigma}=34\%$
$\bar{\sigma}=35\%$
$\bar{\sigma}=50\%$
$\bar{\sigma}=38\%$
$\bar{\sigma}=29\%$
$\bar{\sigma}=27\%$
2016
2017
2018
2019
2020
2021
2022
2023
2024
Year
Shaded regions: major market events
NGX bear market (2016)
Global rate hikes (2022)
Unconditional mean volatility
COVID-19 shock (2020)
Naira devaluation (2023–24)